



KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI

EXECUTIVE SUMMARY

Lead AI Engineer & Data Architect specializing in sovereign autonomous agents, Neo4j GraphRAG, and deterministic civil engineering software. Founder of Archi Cam AI (official applicant for Google Africa Applied AI Lab), bridging multi-agent orchestration with strict mathematical compliance (BAEL 91, ICAO Annex 17) and production MLOps.

FLAGSHIP AI PROJECTS

Archi Cam AI – *Agentic AI & 5D BIM SaaS*

- ▶ Official candidate for Google Africa Applied AI Lab (Accra, Ghana). Sovereign 5D BIM & BOQ estimation platform.
- ▶ Combines local Google Gemma 4 12B QAT, Gemini 1.5 Pro, and deterministic Python Sandbox (IfcOpenShell, BAEL 91).
- ▶ Automates 6-sheet NDA Excel BOQs in <45s (-99.2% time) and synthesizes 3D renders via Imagen 3.0 + ControlNet.

Sovereign.BI Agentic – *Enterprise Security & Agentic BI*

- ▶ Sovereign natural language query engine for complex enterprise SQL & Graph databases.
- ▶ TypeScript Orchestrator architecture, Neo4j N10S (GraphRAG), and FastAPI/PostgreSQL backend.
- ▶ Enforces dynamic ABAC guardrails and anti-hallucination explainability via SHAP Sentinel.

Dataset Automator & VigieSahel – *MLOps Pipeline & Climate AI Impact*

- ▶ Dataset Automator: Time-series evaluation RAG pipeline (Neo4j, MLflow tracking, Firebase Genkit).
- ▶ VigieSahel: Predictive ML platform for agro-climatic sowing optimization & health outbreak tracking ($R^2 > 0.94$).

PROFESSIONAL EXPERIENCE

AI Lead & Data Science Consultant (2025 – Present)

Independent Projects & Enterprises | Douala, CM

- ▶ Guiding enterprises in deploying sovereign local AI agents and private data processing pipelines.
- ▶ Exploratory data analysis and preprocessing of high-dimensional complex datasets.
- ▶ Knowledge graph modeling (Neo4j Cypher) and end-to-end GraphRAG pipeline development.
- ▶ Designing PostgreSQL relational schema and interactive executive reporting dashboards.

Aviation Security Officer (AVSEC) (2018 – Present)

CCAA (Cameroon Civil Aviation Authority)

- ▶ Critical threat assessment, security inspections, and access control management (ICAO Annex 17).
- ▶ Drafting regulatory security compliance reports and coordinating operational field emergency response.

EDUCATION & CERTIFICATIONS

Master of Science in Applied Artificial Intelligence (2025 – 2027) [In Progress]

University of Ngaoundéré | Graph Modeling (Neo4j), MLOps, Prompt Engineering & LLM Architecture

Bachelor of Science in Civil Engineering (Building Option) (2015 – 2016)

ISTDI / IUC Douala | Structural Calculations (BAEL 91), Quantity Surveying & Construction Project Management

CONTACT

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AI & LLM STACK

Google Gemma 4 ■■■■■■
Gemini 1.5 Pro ■■■■■■
CrewAI/LangGraph ■■■■■■
Firebase Genkit ■■■■■■
RAG / GraphRAG ■■■■■■

DATA & GRAPHS

Neo4j / Cypher ■■■■■■
PostgreSQL / SQL ■■■■■■
Supabase ■■■■■■
Firebase Emulator ■■■■■■
Pandas / NumPy ■■■■■■

DEV & MLOPS

Python (BAEL 91) ■■■■■■
IfcOpenShell (5D) ■■■■■■
FastAPI / Next.js ■■■■■■
MLflow (MLOps) ■■■■■■
Streamlit / Docker ■■■■■■

LANGUAGES

French (Native)
English (Professional / Tech)

KEY ASSETS

- ◆ Dual Competence AI & Civil Eng.
- ◆ Aviation Security & Risk (AVSEC)
- ◆ Strict Math Guardrails & 0 Hallucination